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Electron-Ion Collider, Brookhaven National Laboratory			
Doc No. EIC-SEG-RSI-009	Author: L. Strange	Effective Date: January 23, 2026	Review Frequency: 1 year
Requirements, Specifications, and Interfaces: Naming Requirements			Revision: 01

Electron-Ion Collider Requirements, Specifications, and Interfaces

Naming Requirements

January 23, 2026

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REVISION HISTORY

Revision #	Effective Date	List of Reviewers	Summary of Change
00	7/26/2023	C. Hetzel, R. Srinivasan, D. Bruno, D. Gassner, K. Kulmatycki, J. Jamilkowski, K. Smith, T. Russo, A. Zaltsman, J. Reich, and K. Wilson	Initial release.
01	1/23/2026		Generalized syntax and removed specific examples. Added Roles and Responsibilities section.

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LIST OF ACRONYMS

BNL	Brookhaven National Laboratory
EIC	Electron-Ion Collider
EPICS	Experimental Physics and Industrial Control System
ESR	Electron Storage Ring
HSR	Hadron Storage Ring
IP	Interaction Point
RHIC	Relativistic Heavy Ion Collider
SBMS	Standards-Based Management System
SEG	Systems Engineering Group

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Naming Requirements

1. PURPOSE AND SCOPE

The purpose of this document is to define the naming convention for components in the Electron-Ion Collider (EIC) facility. Names will serve as unique identifiers for EIC equipment. Names will be used for interface identification, installation tasks, status tracking, and related documentation across the project.

Naming is managed by the Systems Engineering Group (SEG). SEG is developing a naming repository to facilitate naming management; SEG will directly manage the approved names until the repository is complete.

The naming syntax requirements defined in this document combined standards set by the SBMS (Standards-Based Management System) naming requirements, EPICS (Experimental Physics and Industrial Control System) character limitations, and RHICs (Relativistic Heavy Ion Collider) naming conventions.

2. ROLES AND RESPONSIBILITIES

It is essential that all contributors understand their roles in defining, applying, and maintaining consistent naming conventions.

2.1. Systems Engineering Group

The Systems Engineering Group is responsible for:

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- Defining and maintaining the official naming syntax and structure.
- Reviewing and approving proposed names to ensure compliance, uniqueness, and system compatibility.
- Managing updates and version control for the naming repository.
- Ensuring names are applied consistently across documentation, databases, and installation tracking systems.
- Resolving conflicts or exceptions to the standard and providing guidance on edge cases.

2.2. Equipment Owner Responsibilities

Equipment owners are responsible for defining and documenting equipment names in accordance with the established naming standard. Once names are defined, the naming convention can be used to identify and maintain parent–child relationships across equipment, systems, and components.

Examples include:

- Interfaces to beamline components
- Physical location and spatial relationships
- Networking and control system integration
- Utility interfaces
- Mechanical interface drawings and connection points
- Cable route paths and terminations

Owners should work with the Systems Engineering Group to ensure compliance with the naming standard and to address any issues.

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3. SYNTAX REQUIREMENTS

The generalized nomenclature syntax is *aa:bb-ddpp.zz_nn-ccss*

Naming Elements	Description	Abbreviation Type	Device Type
aa	area	from naming repository	beamline or non-beamline
bb	specific area	from naming repository+ number	non-beamline only
dd	device	from naming repository	beamline or non-beamline
pp	position	number	beamline or non-beamline
zz	secondary position	number	non-beamline only
nn	appended number	number	beamline or non-beamline
cc	controller device	from naming repository	beamline only
ss	signal classification	from naming repository	beamline or non-beamline

3.1. Lattice Device Syntax Requirements

For beamline components and directly controlling/reading devices. The lattice device nomenclature syntax is *aa-ddpp_nn-ccss*

3.1.1. Area (aa)

The area is defined by the machine prefix.

3.1.2. Device (dd)

The device is defined by the component function.

3.1.3. Position (pp)

The position is defined by the S-coordinate distance (in meters) from the machine prefix start.

3.1.4. Append Number (nn)

An append number is applied when multiple instances of the same device type are present within a single meter section. This number indicates the sequential occurrence of the device within that section.

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3.1.5. Controller (cc)

The controller is applied to identify devices that perform direct control or readback functions.

3.1.6. Signal (ss)

The signal is applied when it is necessary to specify a signal type.

3.2. Non-Lattice Device Syntax Requirements

Naming for non-lattice devices is based on their physical location. The non-lattice device nomenclature syntax is *aa:bb-ddpp.zz_nn-ss*

3.2.1. Area (aa) and Specific Area (bb)

The area is defined by the device's installation location. Devices installed within the tunnel shall use the predefined machine prefix, while those located in buildings shall use the corresponding building abbreviation as defined in the approved naming repository.

A specific area is applied to define a distinct location within an area.

3.2.2. Device (dd)

The device is defined by the component function..

3.2.3. Position (pp) and Secondary Position (zz)

The position is applied when more than one device type is located within the same area or specific area.

The secondary position is applied when the device type can have both a horizontal and vertical position.

3.2.4. Append Number (nn)

The append number is applied to identify connection points on a device.

3.2.5. Signal (ss)

The signal is applied when it is necessary to specify a signal type.